

Simple Low-Power Audio Amplifier

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The small-signal amplifier is generally referred to as a voltage amplifier because it usually converts a small input voltage into a much larger output voltage. The audio power amplifier works on the basic principle of converting low-power audio signal to a suitable level to be delivered to the load.

This low-power amplifier circuit is useful for the amplification of sound from small-signal devices such as mobile phones, laptops or desktops.

Circuit and working

As shown in Fig. 1, this circuit is built around a step-down transformer (X1), bridge rectifier BR1, regulators 7809 (IC1) and 7909 (IC2), dual op-amp TL072 (IC3), low-power amplifier LM386 (IC4) and some other components.

The circuit can be divided into two sections—dual power supply section and amplifier section. The dual power supply section is built around step-down transformer X1 (230V AC primary to 12V-0-12V, 1A secondary) and two voltage regulators 7809 and 7909. IC 7809 is a positive voltage regu-

lator, while 7909 is a negative voltage regulator. Diodes D1 and D2 are used to protect IC1 and IC2 against reverse voltages from capacitors connected to the regulators. These regulators provide $\pm 9V$ regulated output for the operation of the circuit.

Use suitable heat sinks with the regulator ICs because they get hot during operation. In case of overheating, there is provision for a thermal shutdown.

The amplifier section is built around TL072 (IC3) and a low-power amplifier LM386. The op-amp A1 of IC3 operates as a low-noise preamplifier. Capacitor C8 is used in order to pass low frequency. The op-amp A2 of IC3 operates as a low-pass filter. For changing the cut-off frequency, you have to change the values of capacitors C11 and C12.

LM386 is a low-power amplifier IC with built-in biasing and inputs that are referred to the ground. It has a gain of 20 and can drive a speaker of 8-ohm impedance.

The circuit is simple to use. You have to simply feed the output from a mobile phone, or any other low-volume device, into RCA1 socket on the PCB. The amplified sound can be



PARTS LIST

Semiconductors:

- IC1 - 7809, +9V regulator
- IC2 - 7909, -9V regulator
- IC3 - TL072 dual op-amp
- IC4 - LM386 low power audio amplifier
- BR1 - 1A bridge rectifier
- D1, D2 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-kilo-ohm
- R2 - 10-kilo-ohm
- R3 - 2.2-kilo-ohm
- R4 - 47-kilo-ohm
- R5-R7 - 33-kilo-ohm
- R8 - 56-kilo-ohm
- R9 - 3.3-ohm
- VR1 - 10-kilo-ohm potentiometer

Capacitors:

- C1, C3 - 1000 μ F, 35V electrolytic
- C2, C4 - 0.33 μ F ceramic disk
- C5, C6 - 1 μ F, 25V electrolytic
- C7, C14 - 220 μ F, 25V electrolytic
- C8 - 4.7 μ F, 25V electrolytic
- C9, C10, C13 - 0.1 μ F ceramic disk
- C11, C12 - 0.01 μ F ceramic disk

Miscellaneous:

- LS1 - 8-ohm, 0.5W speaker
- RCA1 - RCA socket
- X1 - 230V AC primary to 12V-0-12V, 1A secondary transformer
- 8-pin IC bases for ICs
- Heat sinks for regulators

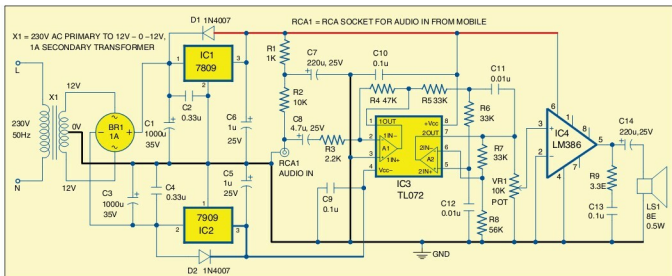


Fig. 1: Circuit diagram of simple low power audio amplifier